

Raphael Perez

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Education

University of Florida – M.S. in Mechanical Engineering Expected December 2026
GPA: 4.00 • Coursework: Advanced Finite Element Methods (Abaqus), Optimal Control, Continuum Mechanics

University of Florida – B.S. in Mechanical Engineering, Minor in Biomechanics, Cum Laude May 2025
GPA: 3.65 • Major GPA: 3.73

Projects

Digest-A-Bull: Cow Rumen Digestion Simulator – Mechanical Design Engineer 2024 – 2025

- Led full design cycle from concept selection (Pugh chart, DFM/DFA) to GD&T drawings per ASME Y14.5 and BOM
- Integrated PT100 RTD, pressure, and methane sensors with controls hardware for closed-loop validation
- Performed stress analysis: hand calculations for bending and fastener loads, validated against SolidWorks FEA
- Implemented PID temperature control ($\pm 0.1^\circ\text{C}$) and proportional motor speed control; validated via NI DAQ testing
- Reduced researcher setup time from multiple days to 79 minutes, increasing experiment throughput by 25%

Engine Gasket Nonlinear FEA and Hyperelastic Calibration – FEA Engineer 2026

- Calibrated Ogden ($R^2=0.9629$) and Mooney-Rivlin hyperelastic models from biaxial stress-strain test data in MATLAB
- Performed nonlinear contact FEA in Abaqus; scripted Python for contact-pressure output; closed 2.22 mm sealing gap
- Conducted mesh convergence study and refined gasket geometry to improve pressure uniformity and sealing

Deep Drawing Sheet Metal Forming Simulation – FEA Engineer 2026

- Selected S4R shell elements and quarter-model symmetry in Abaqus to capture sheet bending, thinning, and springback
- Structured 4-step FEA to prevent convergence failure: binder seating, binder force, 70 mm punch stroke, springback
- Compared 8.8 kN vs 1.25 kN binder force: thinning of 11.22% vs 2.11% and peak punch force of 3161 N vs 648 N

Automatic Fall-Prevention Walker – Mechanical Design Engineer 2023

- Designed structural frame and fall-prevention mechanism in Fusion 360; produced GD&T drawing package per ASME Y14.5
- Performed ANSYS R22.2 FEA under 294.7-lbf combined loading; identified upper frame ($FOS = 1.00$) as critical region
- Applied FMEA and root cause analysis to select 6061-T6 aluminum; improved frame stability 20%

Optimal Control of Nonlinear Systems – Controls and Analysis Engineer 2026

- Applied Pontryagin minimum principle to analytically solve a hyper-sensitive optimal control problem in closed form
- Implemented indirect multiple shooting in MATLAB; achieved terminal state error below $4e-10$ across 6 time horizons
- Applied direct shooting in MATLAB to a robot arm minimum-time problem; reduced transfer time from 12.14 s to 9.63 s

Experience

Propulsion Analyst Intern, Rocket Lab USA • Long Beach, CA August 2026

AI Search Quality Evaluator, Welocalize Inc • Miami, FL June 2024 – July 2026

- Evaluated AI search outputs against structured relevance guidelines, improving result quality metrics by 15%
- Analyzed large query datasets to identify recurring failure modes, edge cases, and ranking inconsistencies

Skills

CAD & Design: SolidWorks, Autodesk Fusion 360, ANSYS, Abaqus, GD&T, DFMA, Bill of Materials, Tolerance Analysis

Simulation & Analysis: Finite Element Analysis, Structural Analysis, Thermal Analysis, FMEA, Root Cause Analysis

Programming & Testing: MATLAB, Python, LabVIEW/Ni DAQ, Functional Testing, PID Control, Microsoft Office Suite